

On Electronegativities of the Rare-Earth Elements Based on Relative Valence Electron Binding Energies

Henning A. Höppe*

Cite This: *Inorg. Chem.* 2026, 65, 1538–1542

Read Online

ACCESS |



Metrics & More



Article Recommendations



Supporting Information

ABSTRACT: The electronegativities of the rare-earth elements are proposed; these use average “relative valence electron binding energies” (ReVEBE) that are deduced from experimental spectroscopic data listed in the NIST Atomic Spectra Database. According to this concept, only the lowest-lying levels of the valence electron configurations are considered. These electronegativities were obtained as estimated binding energies based on a modified concept by Rahm, Zeng, and Hoffmann of ground-state average valence electron binding energies for the main group and transition metal elements. The modification presented here enables determination of the electronegativities of the rare-earth elements. The obtained values are in very good agreement with those proposed by Allen, Pauling, and Mulliken, as well as the optical electronegativities of the lanthanoid elements. This is remarkable since these concepts relate to different physical principles.



INTRODUCTION

Chemically speaking, the most important figure of an element is its electronegativity. Very similar to Avogadro's idea of an “electrical heterogeneity” of 1809,¹ the Swedish chemist Berzelius mentioned the concept of “electronegative” and “electropositive” bodies reacting with each other—to suggest a driving force of chemical reactions.² In 1932, Pauling reported a thorough investigation on bond energies of several compounds; bond energies of covalent bonds between two atoms A and B are additive regarding the arithmetic—later the geometric—mean of A–A and B–B bond energies plus a contribution Δ' , the polar component; here, Δ' represents the squared difference of the electronegativities of A and B, i.e., χ_A and χ_B .^{3–5} These electronegativities describe the atoms' power to attract electrons, thus determining their bonding behavior within the van Arkel–Ketelaar triangle^{6,7} (metallic, ionic, and covalent) and thus implicitly their chemical behavior. In this context, the valence electrons are the decisive ones. Table 1 lists the Pauling electronegativity figures of selected elements. Later, several concepts were suggested, like that of Mulliken's absolute electronegativity based on the arithmetic mean of the respective ionization potential and electron affinity of an atom, physicochemically different parameters than those used by Pauling.^{8,9} In this contribution, the recently published updated values by Cárdenas et al. will be used for comparison.¹⁰

In 1989, Allen proposed the meanwhile famous spectroscopic electronegativity χ_{spec} based on averaged spectroscopically measured energies of valence electrons, based on a further different physicochemical parameter.^{16–18} These energies represent the mean configuration energies of the valence

Table 1. Electronegativity Values of Selected Elements According to Pauling,^{5,11} Allen,^{12,13} Rahm/Zeng/Hoffmann¹⁴ (χ_{RZH} Scaled to Allen's Scale in Pauling Units, See Text), and Mulliken (Updated Values by ref 10) in Pauling Units (χ_{abs} Was Transformed According to ref 15 and Scaled to Allen's Hydrogen Figure (See Text))

	χ_{P}	χ_{Allen}	χ_{RZH}	χ_{abs}
hydrogen	2.20	2.300	2.30	2.30
nitrogen	3.04	3.066	2.86	2.32
oxygen	3.44	3.612	3.14	2.40
fluorine	3.98	4.193	3.94	3.13
scandium	1.36	1.19	1.2	1.18
yttrium	1.22	1.12	1.1	1.15
lanthanum	1.10		1.0	1.07
europium			1.6	1.03
gadolinium	1.20		2.3	1.11
ytterbium			1.7	1.10
lutetium	1.0	1.09	1.1	1.02

electrons. This was the first approach to actually link experimental spectroscopic data of elements to their electronegativity. Initially, Allen focused on main group elements; later,

Received: October 31, 2025

Revised: December 19, 2025

Accepted: December 30, 2025

Published: January 7, 2026



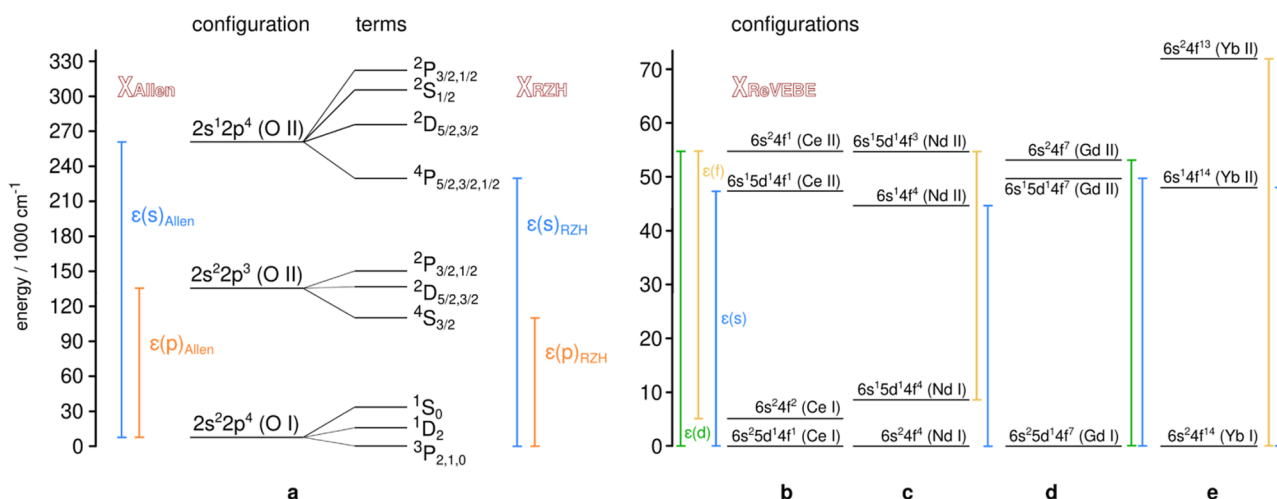


Figure 1. (a) Allen's concept of configuration energies on the example of oxygen with the corrected energies of this work, compared to that of binding energies by Rahm/Zeng/Hoffmann (also (b) and (d)); (c,e) illustration of the modified approach in this work.

he also included the outer transition metals into this concept.^{12,13} He determined the $(2J + 1)$ - and $(2S + 1)(2L + 1)$ -weighted s -, p -, and d -orbital energies of the relevant electronic configurations, the configuration energies, as depicted in Figure 1a. Considering the shown example of oxygen, the configuration energy of a p electron is the energy difference between the $2s^2p^4$ and the $2s^2p^3$ configurations, and the configuration energy of an s electron is the energy difference between the $2s^2p^4$ and the $2s^1p^4$ configurations. All of the relevant figures had been determined already and are still available as spectra "O I" and "O II" in the NIST Atomic Spectra Database¹⁹ for the neutral and singly oxidized oxygen atoms. Based on these configuration energies ϵ_s and ϵ_p , the mean configuration energy of a main group atom's valence electrons amounts to

$$\chi_{\text{spec}} = \frac{u \cdot \epsilon(s) + v \cdot \epsilon(p)}{u + v} \quad (1)$$

Regarding the outer transition metals, Allen restricted the relevance of valence electrons to the maximum oxidation state of the respective metal atoms. u and v represent, therefore, the number of relevant respective valence electrons.¹³ Since he included all possible terms of the valence electrons of an element to get the averaged energy—also the highly excited ones—this approach is limited to the main group and outer transition metals. This approach failed, though, for some outer and for all inner transition metals—the lanthanoid elements—which are part of the rare-earth elements, due to the huge number of possible terms; their states overlap in many cases energetically and thus cannot be discriminated. For each observed transition, only the major components can be assigned reliably, but nevertheless, the ground energies of many configurations are clear.

In 2018, Rahm, Zeng, and Hoffmann approached this topic and suggested employing only the ground-state energies of the relevant electronic configurations to determine the valence electron binding energies.¹⁴ Such a restriction sounds reasonable because electronegativity is normally employed for systems at temperatures far below the high energies, which the excited terms require to be relevant. The ground-state energies are certainly somewhat lower than the total averaged ones, as illustrated in Figure 1a. Thus, this restriction led to slightly more

electropositive values for all elements, as shown for selected elements in Table 1. They determined the respective averaged binding energies for the main group and the majority of the transition metal elements, solely based on these spectroscopic data. Unfortunately, for several elements, among these the lanthanoid elements, they did not use spectroscopically determined 4f, 5d, and 6s binding energies but theoretically calculated ones as they cannot be deduced from the desired ground-state configurations in most cases—this is not a criticism. So, there arose a strong motivation to address this problem, that not all relevant elements feature spectroscopically determined electronegativity values. In rare-earth chemistry, this is a long missing link.

In this contribution, the author proposes spectroscopically obtained electronegativities of the rare-earth elements based on average relative valence electron binding energies (ReVEBE). This represents a modified concept of Rahm, Zeng, and Hoffmann, which is based on ground-state average valence electron binding energies and indicated the RZH concept here.¹⁴ The author will suggest a possible solution to obtain spectroscopically recorded binding energies for the lanthanoid elements. The new figures will be given as binding energies in wavenumbers, eV, and in Pauling units as suggested by Allen.¹⁶ Moreover, the author will discuss an at-first-glance astonishing link to another spectroscopic property, the relative energies of ligand-to-lanthanoid charge-transfer transitions, to prove the validity of the ReVEBE concept.

RESULTS

Electronegativities of the Rare-Earth Elements Based on Average Valence Electron Binding Energies

Based on the knowledge of rare-earth chemistry, the f -levels certainly determine the optical and magnetic properties of the elements as well as the ions, but only a few contribute to their chemistry as they are hidden deeply inside the atoms.²⁰ The electronegativities in this contribution are deduced from the average "relative valence electron binding energies" (ReVEBE) determined following these rules:

- Only valence electrons of the outermost shell plus further of lower shells up to the highest experimentally observed oxidation state of each element are considered, as suggested by Allen;¹³ indeed, all f electrons might be

Table 2. Averaged Spectroscopic s-, d-, and f-Level Binding Energies $\epsilon(s)$, $\epsilon(d)$, and $\epsilon(f)$ (*Mean of Neighboring Elements, See Text), the Respective Electron Counts x , y , and z According to Eq 2 as well as the Resulting $\bar{\chi}$ of This Work in Wavenumbers, eV, and Pauling Units (PU, See Text) for Comparison with the Figures of Rahm/Zeng/Hoffmann, and the Absolute and Pauling Electronegativities in PU

	$\epsilon(s)/\text{cm}^{-1}$	x	$\epsilon(d)/\text{cm}^{-1}$	y	$\epsilon(f)/\text{cm}^{-1}$	z	$\bar{\chi}_{\text{ReVEBE}}/\text{cm}^{-1}$	$\bar{\chi}_{\text{ReVEBE}}/\text{eV}$ (this work)	$\bar{\chi}_{\text{RZH}}/\text{eV}$ (ref 14)	$\bar{\chi}_{\text{ReVEBE}}/\text{PU}$ (this work)	$\chi_{\text{abs}}/\text{PU}$ (ref 10)	$\chi_{\text{P}}/\text{PU}$
Ca	49,306	2					49,306	6.12	6.1	1.03	1.08	1.00
Sc	55,463	2	60,266	1			57,064	7.08	7.0	1.20	1.18	1.36
Y	50,986	2	50,146	1			50,706	6.29	6.3	1.06	1.15	1.22
La	46,876	2	52,376	1			48,709	6.04	6.0	1.02	1.07	1.10
Ce	47,268	2	54,451	1	49,688	1	49,562	6.15	7.3	1.04	1.09	1.12
Pr	44,120	2	36,040	0	43,929	2	44,025	5.46	6.7	0.92	1.13	1.13
Nd	44,560	2	36,085	0	46,139	1	45,086	5.59	7.2	0.95	1.01	1.14
Pm	45,021	2		0	51,183*	1	47,075	5.84	7.4	0.99	1.01	
Sm	45,520	2	34,719	0	56,227	1	49,089	6.09	8.3	1.03	1.03	1.17
Eu	45,735	2	32,811	0	63,000	1	51,490	6.39	9.4	1.08	1.03	
Gd	49,601	2	53,045	1	42,071	0	50,749	6.30	13.8	1.06	1.11	1.20
Tb	47,295	2	52,907	0	53,193	2	50,244	6.23	7.7	1.05	1.11	
Dy	47,902	2	52,672	0	60,238	1	52,014	6.45	8.4	1.09	1.11	1.22
Ho	48,567	2	54,551	0	62,930	1	53,355	6.62	8.3	1.12	1.12	1.23
Er	49,262	2	48,910	0	56,087	1	51,537	6.39	7.6	1.08	1.13	1.24
Tm	49,881	2	45,531	0	58,651	1	52,804	6.55	9.0	1.11	1.09	1.25
Yb	50,443	2	48,674	0	71,862	1	57,583	7.14	10.2	1.21	1.10	
Lu	55,559	2	43,763	1		0	51,627	6.40	6.4	1.08	1.02	1.0

included—not dramatically changing the results—but this seems chemically not plausible; in the case of the rare-earth elements, only for cerium, praseodymium, and terbium were the pure tetrafluorides reported experimentally yet.

- The actual valence electrons are identified by the configurations of the ionized species according to the spectra I, II, III, and—if applicable—IV of the rare-earth elements; this is relevant, as it makes a difference whether, for instance, a 6s electron is removed from a configuration $6s^2 5d^1 4f^z$ or $6s^2 5d^0 4f^{z+1}$ since the shielding of the removed valence electron is different.
- The energetically lowest-lying electronic configurations describing the removal of a single respective valence electron listed in the NIST database are used as depicted in Figure 1b–e, depending on the electron configurations of the respective rare-earth atom; for details of the calculation used for each element, the author refers to Supporting Information II.
- The average relative valence electron binding energies are obtained from the energy differences between these configurations.

Therefore, the author proposes deducing the electronegativities via the equation

$$\bar{\chi}_{\text{ReVEBE}} = \frac{x \cdot \epsilon(s) + y \cdot \epsilon(d) + z \cdot \epsilon(f)}{x + y + z} \quad (2)$$

The predominant oxidation state of the rare-earth elements is three. For scandium, yttrium, lanthanum, gadolinium, and lutetium, these three ionized electrons are the outermost single d and the two s electrons. In the case of cerium, the remaining 4f electron has to be included since its maximum oxidation state is four. Considering the remaining elements, mainly a single 4f electron together with the two 6s electrons are ionized; exceptions are praseodymium and terbium with maximum oxidation state four, where two 4f electrons were included. As illustrated in Figures 1b and 1d, the binding energies of the s

electrons were obtained from the energy differences between the configurations $(n)s^2(n-1)d^1$ and $(n)s^1(n-1)d^1$ for Sc, Y, La, Ce, Gd, and Lu; for the remaining elements, these energies were determined from the configurations $6s^2 4f^z$ and $6s^1 4f^z$ as shown in Figures 1c and 1e. Similar approaches were used for the d electron energies, employing the configurations $(n)s^2(n-1)d^1$ and $(n)s^2$ with a constant 4f configuration, where applicable. These obtained s and d energies resemble the values reported by Rahm/Zeng/Hoffmann.¹⁴ The experimental binding energies of the 4f electrons were determined from the configurations $6s^2 4f^z$ and $6s^2 4f^{z-1}$ (Figures 1b and 1e) or $6s^1 5d^1 4f^z$ and $6s^1 5d^1 4f^{z-1}$ (Figure 1c). Table 2 lists the average relative valence electron binding energies, the number of included 6s, 5d, and 4f valence electrons, and the yielded electronegativity figures $\bar{\chi}_{\text{ReVEBE}}$ and compares these results to the aforementioned electronegativity scales. The author added the values of calcium for comparison, as this chemically resembles the metals quite well. These, as well as the details of the calculations, like the configurations used, are attached to this contribution as Supporting Information II. For promethium, only the s electron binding energy could be deduced from NIST data, which approximately matches the mean of the s electron binding energies of its neighbors; the f electron energy might therefore be extrapolated as the mean energies of the neighboring elements neodymium and samarium. Solely, the energy of the $6s^2 4f^{10}$ configuration of holmium had to be confirmed by recently calculated data.²¹

DISCUSSION

The obtained electronegativities of the rare-earth elements based on average relative valence electron binding energies (ReVEBE) resemble more or less the figures already proposed by Pauling for most of the elements. The same holds for those of the Mulliken so-called absolute electronegativity, revised by Cárdenas et al.,¹⁰ and transformed on the Pauling scale employing the fit function by Herrick¹⁵ in Table 2. For practical reasons, the author lists in this contribution the energy values of

$\bar{\chi}_{\text{ReVEBE}}$ as well as the figures in Pauling units, as suggested by Allen.¹⁶ The coincidence of all three scales is remarkable since the three concepts refer to different physical data of an element. Pauling related the squared differences of χ_p of the bonding partners to the electrostatic contribution to the bond energy, i.e., thermochemical data. Mulliken used the electronic chemical potentials, employing the ionization energies and electron affinities. Allen's approach and that of Rahm/Zeng/Hoffmann, which is the base of the modified concept used here, are fairly similar and use the configuration energies that are accessible via spectroscopy. Apparently, the calculated binding energies of the lanthanoid elements in ref 14 are remarkably different from the $\bar{\chi}_{\text{ReVEBE}}$ figures determined spectroscopically here.

Considering the differences between the suggested electronegativities by Rahm et al. (Table 2)—based on theoretically calculated 4f, 5d, and 6s binding energies—the most prominent differences are the electronegativity of gadolinium and, in general, the significantly higher figures for the lanthanoid elements. According to these results, gadolinium should be rather electronegative, comparable to mercury, and even more electronegative than gold, for instance, as depicted in Figure 2.

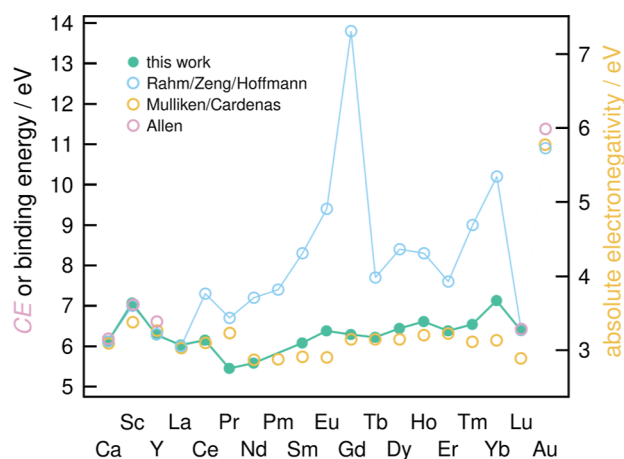


Figure 2. Comparison of the obtained binding energies of this work (full green spheres) with those of Rahm/Zeng/Hoffmann (blue circles), Allen (red circles, all left ordinate), and the absolute electronegativities (orange circles, right ordinate); data: refs 10 and 12–14.

Hence, gadolinium would be by far the most electronegative element of the series, and chemists might expect gadolinium to prefer lower oxidation states. For divalent gadolinium, to the best of the author's knowledge, no compound is known where indeed divalent gadolinium is contained—instead, trivalent ions and an itinerant electron are observed, like in GdI_2 .²² Within the $\bar{\chi}_{\text{ReVEBE}}$ scale, europium and ytterbium feature the highest electronegativities of the early and late lanthanoid elements, respectively. The reason for these differences lies not in whether the f electrons are considered to be part of the valence shell—which might be relevant in cases where they indeed behave as valence electrons. Interestingly, the general trend of the rare-earth electronegativities of Rahm/Zeng/Hoffmann is the same as in this work—with the exception of gadolinium, where the half-occupied 4f-shell lies highly stabilized deep inside the atoms. Hence, in this contribution, these electrons are not included. But even if they were included, this would not change gadolinium's electronegativity dramatically. Figure S2 in Supporting Information I illustrates the effect of including all

4f electrons—it emphasizes the same aforementioned trend. Furthermore, this observed trend within the $\bar{\chi}_{\text{ReVEBE}}$ scale matches nicely the trend of the “optical electronegativities” of the lanthanoid elements, a concept suggested by Jørgensen.²³ He found that the energies of charge-transfer transitions between ligands and coordinated metal ions roughly match their electronegativity difference according to

$$E_{X \leftrightarrow M}^{\text{CT}} \approx 30,000 \text{ cm}^{-1} \cdot (\chi_{\text{opt}}(\text{X}) - \chi_{\text{opt}}(\text{M})) \quad (3)$$

Dorenbos systematically investigated the charge-transfer energies of transitions between trivalent rare-earth ions RE^{3+} and coordinating atoms. He identified a general pattern regarding the charge-transfer energies of the whole series of RE^{3+} . Although the charge-transfer energies certainly depend strongly on the host structure, the relative differences of charge-transfer energies within the series of RE^{3+} in the same host structure are pretty much the same. Thereby, always the compound hosting trivalent europium ions featured the lowest charge-transfer energy.^{24–26} Figure S3 in Supporting Information I shows the general charge-transfer energy differences of the trivalent lanthanoid ions deduced from a huge number of host structures.^{24,25} Considering eq 3, these relative charge-transfer energies to arbitrary ligands X are proportional to the respective rare-earth element's electronegativity, where europium is the most electronegative and lanthanum and gadolinium are the most electropositive elements of the series.

The pattern of these optical electronegativities matches the pattern of the electronegativities $\bar{\chi}_{\text{ReVEBE}}$ and is thus actually the proof of principle. This coincidence was at first glance astonishing, as the electronic configurations of the trivalent rare-earth elements were not used but the energies of the valence electrons starting from a neutral atom were. As a chemist, the author is fascinated when data of an element fit other data of this element determined by different physical means. Moreover, the systematic behavior of the trivalent lanthanoid charge transfer energies can be rationalized using the electronegativity values of this contribution.

CONCLUSIONS

In this contribution, the author proposed electronegativities $\bar{\chi}_{\text{ReVEBE}}$ of the rare-earth elements based on average relative valence electron binding energies (ReVEBE). Now, a further increasingly relevant group of elements within the periodic table can be assigned spectroscopically determined electronegativities. The trend within the suggested electronegativities matches the trend of differences of charge-transfer energies between rare-earth ions and ligands, which can now be rationalized based on $\bar{\chi}_{\text{ReVEBE}}$. Moreover, the $\bar{\chi}_{\text{ReVEBE}}$ coincide with the few values suggested by Pauling and Allen previously, as well as with the absolute electronegativities by Mulliken.

EXPERIMENTAL SECTION

The electronegativities were calculated employing the XLSX file attached to this contribution as Supporting Information II. In the worksheet ReVEBE, the results according to eq 2 are summarized for the rare-earth and several other elements. It lists the atomic number, the ReVEBE energies of s, p, d, and f electrons in wavenumbers, Rydbergs, and electron volts, and finally $\bar{\chi}_{\text{ReVEBE}}$ in Pauling Units (PU); the relative Pauling scale is based on hydrogen, whose configuration energy is 1 Ry, as suggested by Allen, yielding $\chi_{\text{Allen}} = 2.30016$ for hydrogen;¹⁶ following this suggestion, also the figures by Rahm/Zeng/Hoffmann, χ_{RZH} , were accordingly scaled in this work for comparison purposes.

The worksheet Rare-Earth ReVEBE Calculation was used to calculate the relative energies of the lanthanoid elements in wavenumbers taken from https://physics.nist.gov/PhysRefData/ASD/levels_form.html. Green, yellow, and violet numbers relate to 6s, 5d, and 4f energies, respectively, and the blue fields relate to the obtained averaged binding energies in wavenumbers, electronvolts, and Pauling units. These latter were for comparison purposes also calculated, including all f electrons. The single orange value is taken from ref 21, and the single yellowish-colored field is the extrapolated value of the 4f energy of promethium as the mean energies of the neighboring elements neodymium and samarium. The work sheets O (Allen's approach) and F (Allen's approach) contain the details of the reproduced calculations of Allen's concept for oxygen and fluorine based on actual NIST data.

■ ASSOCIATED CONTENT

SI Supporting Information

The following files are available free of charge: The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.5c05115>.

Comparison of electronegativities deduced by including the relevant and all 4f electrons and comparison with charge-transfer energies of the rare-earth elements^{25,26} (PDF)

Calculation of the electronegativity values of this work based on the NIST data, including the χ_{ReVEBE} for the elements 1 through 72, which mostly coincide with the figures of Rahm, Zeng, and Hoffmann, excluding most of the outer transition metals treated in a separate contribution (XLSX)

■ AUTHOR INFORMATION

Corresponding Author

Henning A. Höpfe – Solid State Chemistry, Institut für Physik, Universität Augsburg, Augsburg D-86159, Germany;

orcid.org/0000-0002-8734-8258; Phone: +49 (0)821

598-3033; Email: henning.hoeppe@uni-a.de

Complete contact information is available at: <https://pubs.acs.org/10.1021/acs.inorgchem.5c05115>

Notes

The author declares no competing financial interest.

■ ACKNOWLEDGMENTS

The author thanks the Deutsche Forschungsgemeinschaft (DFG) for financial support under the project no. 556118528. The author also would like to thank the reviewers for their helpful and constructive comments.

■ REFERENCES

- (1) Avogadro, A. Idées sur l'acidité et l'alcalinité. *Journal de physique, de chimie, d'histoire naturelle et des arts* **1809**, 69, 142–148.
- (2) Berzelius, J. J. An Explanatory Statement of the Notions or Principles upon which the Systematic Arrangement is Founded, which was Adopted as the Basis of an Essay on Chemical Nomenclature. *A Journal of Natural Philosophy, Chemistry, and the Arts* **1813**, 142–146, 153–166.
- (3) Pauling, L. The Nature of the Chemical Bond. IV. The Energy of Single Bonds and the Relative Electronegativity of Atoms. *J. Am. Chem. Soc.* **1932**, 54, 3570–3582.
- (4) Pauling, L.; Sherman, J. A. Quantitative Discussion of Bond Orbitals. *J. Am. Chem. Soc.* **1937**, 59, 1450–1456.

- (5) Pauling, L. *The Nature of the Chemical Bond and the Structure of Molecules and Crystals: An Introduction to Modern Structural Chemistry*, 3rd ed.; Cornell University; Cornell University Press, 1960.
- (6) van Arkel, A. E. *Moleculen En Kristallen*; W.P. Van Stockum and Zoon; Den Haag, 1941.
- (7) Ketelaar, J. A. A. *Chemical Constitution*, 2nd ed.; Elsevier, Amsterdam, 1958.
- (8) Mulliken, R. S. A New Electroaffinity Scale; Together with Data on Valence States and on Valence Ionization Potentials and Electron Affinities. *J. Chem. Phys.* **1934**, 2, 782–793.
- (9) Mulliken, R. S. Electronic Structures of Molecules XI. Electroaffinity, Molecular Orbitals and Dipole Moments. *J. Chem. Phys.* **1935**, 3, 573–585.
- (10) Cárdenas, C.; Heidar-Zadeh, F.; Ayers, P. W. Benchmark values of chemical potential and chemical hardness for atoms and atomic ions (including unstable anions) from the energies of isoelectronic series. *Phys. Chem. Chem. Phys.* **2016**, 18, 25721–25734.
- (11) Allred, A. Electronegativity values from thermochemical data. *J. Inorg. Nucl. Chem.* **1961**, 17, 215–221.
- (12) Mann, J. B.; Meek, T. L.; Allen, L. C. Configuration Energies of the Main Group Elements. *J. Am. Chem. Soc.* **2000**, 122, 2780–2783.
- (13) Mann, J. B.; Meek, T. L.; Knight, E. T.; Capitani, J. F.; Allen, L. C. Configuration Energies of the d-Block Elements. *J. Am. Chem. Soc.* **2000**, 122, 5132–5137.
- (14) Rahm, M.; Zeng, T.; Hoffmann, R. Electronegativity Seen as the Ground-State Average Valence Electron Binding Energy. *J. Am. Chem. Soc.* **2019**, 141, 342–351.
- (15) Herrick, D. R. Connecting Pauling and Mulliken Electronegativities. *J. Chem. Theory Comput.* **2005**, 1, 255–260.
- (16) Allen, L. C. Electronegativity is the average one-electron energy of the valence-shell electrons in ground-state free atoms. *J. Am. Chem. Soc.* **1989**, 111, 9003–9014.
- (17) Allen, L. C. Extension and completion of the periodic table. *J. Am. Chem. Soc.* **1992**, 114, 1510–1511.
- (18) Allen, L. C. Chemistry and electronegativity. *Int. J. Quantum Chem.* **1994**, 49, 253–277.
- (19) Kramida, A.; Ralchenko, Y. *NIST Atomic Spectra Database, NIST Standard Reference Database 78*; Version 5.12, 2024.
- (20) Höpfe, H. *Rare-Earth Elements: Solid State Materials: Chemical, Optical and Magnetic Properties*; De Gruyter, 2024.
- (21) Radziūte, L.; Gaigalas, G.; Kato, D.; Rynkun, P.; Tanaka, M. Extended Calculations of Energy Levels and Transition Rates for Singly Ionized Lanthanide Elements. II. Tb–Yb. *Astrophys. J. Suppl. Ser.* **2021**, 257, 29.
- (22) Felser, C.; Ahn, K.; Kremer, R.; Seshadri, R.; Simon, A. Giant Negative Magnetoresistance in GdL₂: Prediction and Realization. *J. Solid State Chem.* **1999**, 147, 19–25.
- (23) Jørgensen, C. K. *Orbitals in Atoms and Molecules*; Academic Press, 1962.
- (24) Dorenbos, P. Systematic behaviour in trivalent lanthanide charge transfer energies. *J. Phys.: Condens. Matter* **2003**, 15, 8417–8434.
- (25) Dorenbos, P. Lanthanide charge transfer energies and related luminescence, charge carrier trapping, and redox phenomena. *J. Alloys Comp.* **2009**, 488, 568–573. Proceedings of the 25th Rare Earth Research Conference, June 22–26, Tuscaloosa, Alabama, {USA}
- (26) Dorenbos, P. The Eu³⁺ charge-transfer energy and the relation with the band gap of compounds. *J. Luminesc.* **2005**, 111, 89–104.